

MD SAMIUL ALAM SUMEL

AI Product Engineer — Port & Logistics Domain — Freelance & Remote

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HOW I WORK

Describe the workflow and where it breaks down — I ask questions until the requirement is clear and written down, using 12+ years of port operations experience to judge what actually fits, then design, build, and deploy the solution: frontend independently, backend/data work with AI-assisted development (Claude Code). For a company or industry, the goal is usually the same: find the manual step in a process that already runs — a paper form, a shared spreadsheet, a hand calculation — and replace it with a deployed tool the whole team can use. Every build gets manually tested against the real business rules and explained back before I call it done. An AI Product Engineer (React/Next.js, Node.js/Express, MongoDB) who worked through a large part of an intensive full-stack curriculum, backed by 3 live web apps built for real daily use — not demos. Available for freelance and contract work worldwide, no time zone constraints.

SERVICES I DELIVER

Full-Stack Web Apps

React/Next.js on the frontend, Node.js/Express + MongoDB on the backend — REST APIs, JWT/Firebase authentication, and Stripe payments when the job needs them. Scope starts from a written, agreed requirement, not a guess.

✓3 apps in daily production use (vanilla JS + Cloudflare Workers); React/Next.js via World Kitchen Atlas, Node/MongoDB backend via AI-assisted development.

Billing & Tracking Dashboards

Custom billing calculators, tracking dashboards, and operational tools — the same kind of tool built to solve real problems at my own workplace, after talking through exactly how the numbers are supposed to work with the people who use them.

✓Built for real daily use, not a demo.

Port & Logistics Workflow Digitisation

Turning paper-based or Excel-based port and logistics workflows — wharfrent billing, overtime payroll, cargo/vehicle tracking — into deployed, team-wide web tools, backed by first-hand operational knowledge.

✓Proven with 3 tools converted from manual/Excel processes.

Manual Process Automation

For any company or industry: map how a process actually runs today, find the step still done by hand, and replace it with a deployed tool the whole team can use, not just a one-off script.

✓portbill, otbill, and carview each replaced a specific manual step and are still in daily use.

REMOTE & FREELANCE READINESS

- No time zone constraints — any schedule, worldwide.
- Self-managed: scopes, builds, and deploys independently, manually verified before every release.
- Async-friendly English, Git/GitHub-based collaboration (branching, PRs).
- Every past project is publicly verifiable before you hire.

FEATURED PROJECTS

Port Billing Calculator — [Live](#) | [GitHub](#)

Problem: Port billing system creates permanent entries — C&F agents had no way to estimate wharfrent charges before committing, causing billing disputes, excessive dwell time, and repeated counter visits.

Real-time advance wharfrent calculator: slab-based charge computation, VAT and levy calculation, inside/outside cargo split, hoisting charge auto-calculation, print-ready A4 output. Actively used by C&F agents at Mongla Port daily.

Tech: JavaScript, HTML/CSS, Cloudflare Pages, PWA, Print Output

Daily Car Balance & Location Tracking System ("carview") — [Live](#) | [GitHub](#)

Problem: Vehicle positions across warehouse, shed, and yard were recorded in one person's personal Excel file — invisible to all other staff, creating bottlenecks and dependency on a single individual.

Offline-first PWA tracking 8 port locations, with Cloudflare Worker + private GitHub repo sync, a Chart.js analytics dashboard (7 charts + KPIs), 13 report sections, and Excel export. Built for Mongla Port Authority's Traffic Department.

Tech: PWA, Cloudflare Worker, Chart.js, GitHub Sync, Excel Export

OT Bill Management System — [Live](#) | [GitHub](#)

Problem: Overtime billing was a fully manual Excel process — multi-step hourly rate calculations and final bill generation done by hand every cycle.

Staff enters an employee profile once; the system generates the complete final OT bill instantly with correct hourly rate, cumulative date-wise OT calculation, and A4 print-ready output. Reduced billing cycle time from hours to minutes.

Tech: JavaScript, Employee DB, Auto-Calc, A4 Print, Cloudflare Pages

Client Intake Form — [Live](#) | [GitHub](#)

Problem: Collecting project requirements from a client over email or chat is unstructured — details get missed, wasting time on both sides.

Single-file, client-side project requirement intake form — structured questions, no backend required, mailto-based report so the completed brief lands directly in an inbox.

Tech: JavaScript, HTML/CSS, Client-Side Only

TECHNICAL SKILLS

Domain & Port Operations: Port billing & tariff systems, wharfrent/VAT/levy calculation, cargo & vessel tracking, C&F agent workflows, customs/regulatory compliance context

Frontend & JavaScript: HTML5 (Semantic), CSS3, Responsive Design, Tailwind CSS, DaisyUI, JavaScript (ES6+), DOM & Events, Fetch & Promises / Async-Await, Array/Object Methods, ES Modules

React & Next.js: Own: React (Hooks, Context), React Router, React Hook Form. AI-assisted: TanStack Query, Axios (incl. Interceptors), Next.js App Router, NextAuth

Backend, Data & Deployment: Own: Git & GitHub (Branching, PRs), deployment (Cloudflare Pages/Firebase/Netlify). AI-assisted: Node.js & Express, REST APIs (CRUD), MongoDB (Atlas), Aggregation Pipeline, JWT Auth, Firebase Auth & Admin SDK, Next.js API Routes, Stripe Payments, Pagination & Search

Linux & Automation (self-directed): Bash Scripting, Linux Administration (Ubuntu), systemd, Cron & Timers, Monitoring & Alerting Scripts

PROFESSIONAL BACKGROUND

12+ years of professional experience at Mongla Port Authority — Senior Outdoor Assistant, Port Operations & IT Systems (Nov 2017 – Present); Junior Outdoor Assistant, Port & Terminal Operations (Nov 2013 – Nov 2017). Wharfrent billing, terminal operations, cargo dwell time management, and C&F agent coordination — a track record of discipline and reliability, not just technical skill.

SELF-DIRECTED STUDY

The Odin Project & Full Stack Open

Aug 2026 – Present

Self-Directed — In Progress

Self-paced study for code comprehension, not a certificate — starting from The Odin Project's Intermediate HTML and CSS course, then Full Stack Open (University of Helsinki): JavaScript fundamentals, Node.js/Express in depth, testing, and Git workflow, built from assignment briefs rather than a course guide.

EDUCATION & LANGUAGES

Education: HSC — Business Studies, Khulna Public College (2009–2011); SSC — Business Studies, Mongla Bandar Secondary School (1999–2009)

Languages: Bengali — Native | English — Professional